

Order-Invariant Real Number Summation

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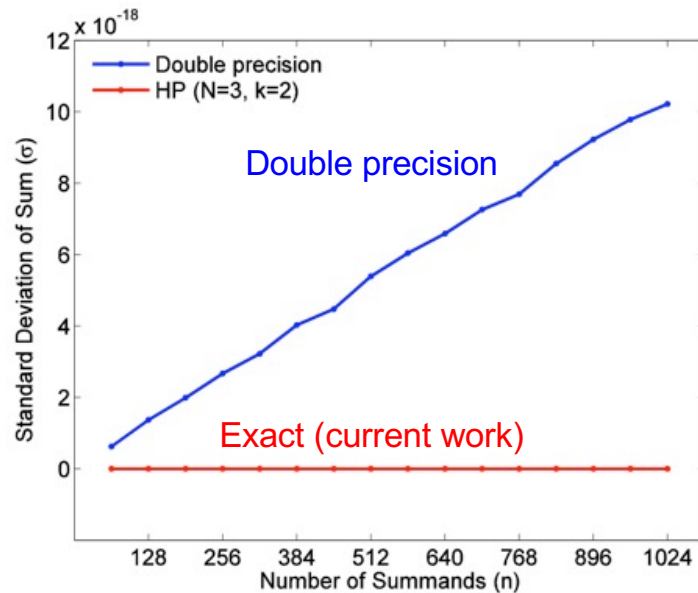
P. E. Small *et al.*, *Proc. IEEE IPDPS*, p. 152 ('16)
<https://aiichironakano.github.io/cs596/Small-OrderInvariantSum-IPDPS16.pdf>



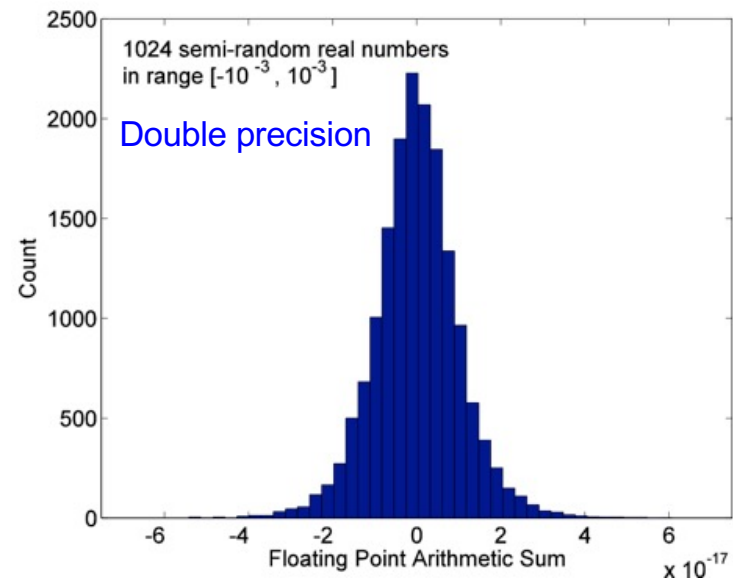
Reproducibility Challenge

- Rounding (truncation) error makes floating-point addition **non-associative**

$$(a + b) + c \neq a + (b + c)$$



Standard deviation of sum with random summation orders



Distribution of sum with random summation orders

- **Finding:** Sum becomes a **random walk** across the space of possible rounding error

Solution: High-Precision (HP) Method

- Propose an extension of the order-invariant, higher-precision intermediate-sum method by Hallberg & Adcroft [*Par. Comput.* **40**, 140 ('14)]
- The proposed variation represents a real number r using a set of N 64-bit unsigned integers, a_i ($i \in [0, N - 1]$)

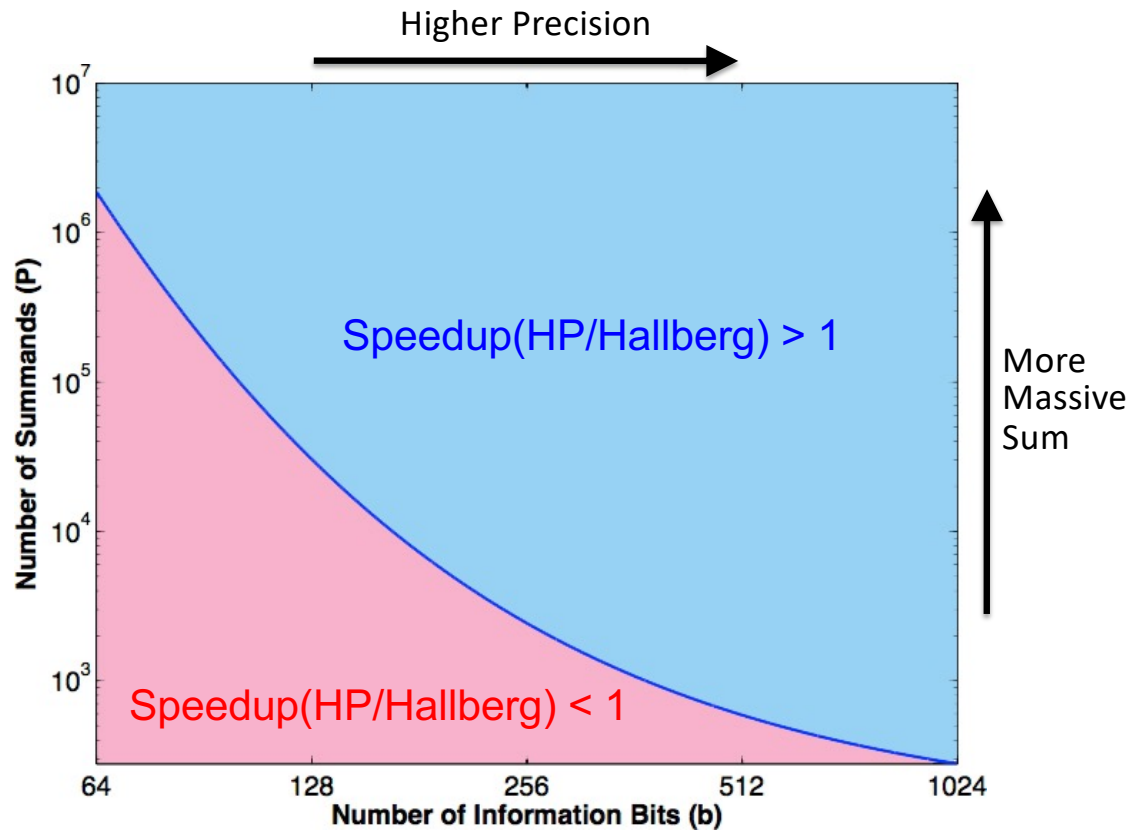
$$r = \sum_{i=0}^{N-1} a_i 2^{64(N-k-i-1)}$$
$$= \overbrace{a_0 2^{64(N-k-1)} + \dots + a_{N-k-1}}^{N-k} + \overbrace{a_{N-k} 2^{-64} + \dots + a_{N-1} 2^{-64k}}^k$$

- k is the number of 64-bit unsigned integers assigned to represent the *fractional* portion of r ($0 \leq k \leq N$), whereas $N-k$ integers represent the *whole-number* component
- Negative number is represented by two's complement in integer representation, using only 1 bit

If you are the first to find the problem, the simplest solution suffices to prove the concept

Performance Projection

- HP sum is faster than Hallberg sum for higher precision & larger numbers of summands



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